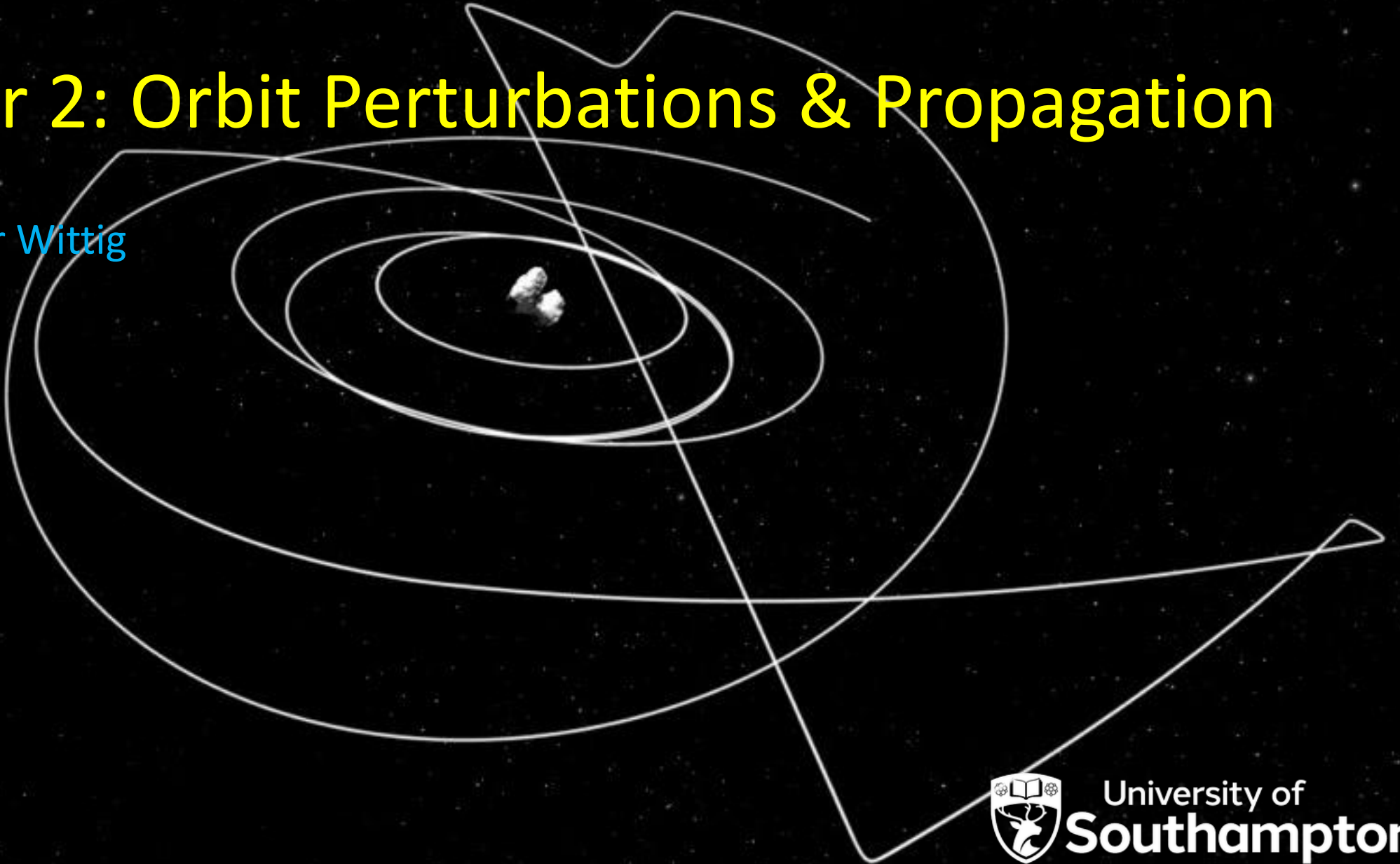


Orbital Mechanics (SESA6076)

Chapter 2: Orbit Perturbations & Propagation

Dr. Alexander Wittig



- Any questions on previous weeks content?

- **Introduction**
- **Chapter 1: In 20.000 km, turn right. Orbital Manoeuvres**
 - Coplanar Manoeuvres
 - Non-Coplanar Transfers
- **Chapter 2: That's mildly disturbing Orbital Perturbations & Propagation**
 - Non-Uniform Gravity Field
 - Third-Body Perturbation
 - Atmospheric Drag
 - Solar Radiation Pressure
 - Orbit Propagation Methods
- **Chapter 3: Is it a plane? Is it a bird? Orbit Observation & Determination**
 - Observation Techniques
 - Preliminary Orbit Determination
- **Chapter 4: Waltzing Matilda Relative Orbital Motion**
 - Relative Orbital Motion
 - Linearization & State Transition Matrix
- **Chapter 5: To boldly go... Interplanetary Trajectories & Optimization**
 - Lambert's Problem
 - Patched Conics
 - Gravity-assisted fly-by trajectories
- **Chapter 6: Torn between two bodies Restricted Three-Body Problem**
 - N-Body Problem
 - Circular Restricted Three-Body Problem
 - Natural Dynamics around Libration Points
- **Chapter 7: You spin me right round Attitude Dynamics**
 - Attitude representation
 - Attitude Dynamics
- **Chapter 8: In the spin cycle Attitude Motion**
 - Torque-Free Motion
 - Attitude Maneuvers
- **Summary and Revision**

- Identify orbit perturbations and their relevance for typical Earth-centric and interplanetary orbits
- Apply different orbit propagation techniques to account for perturbations
- Design frozen orbits for common perturbations

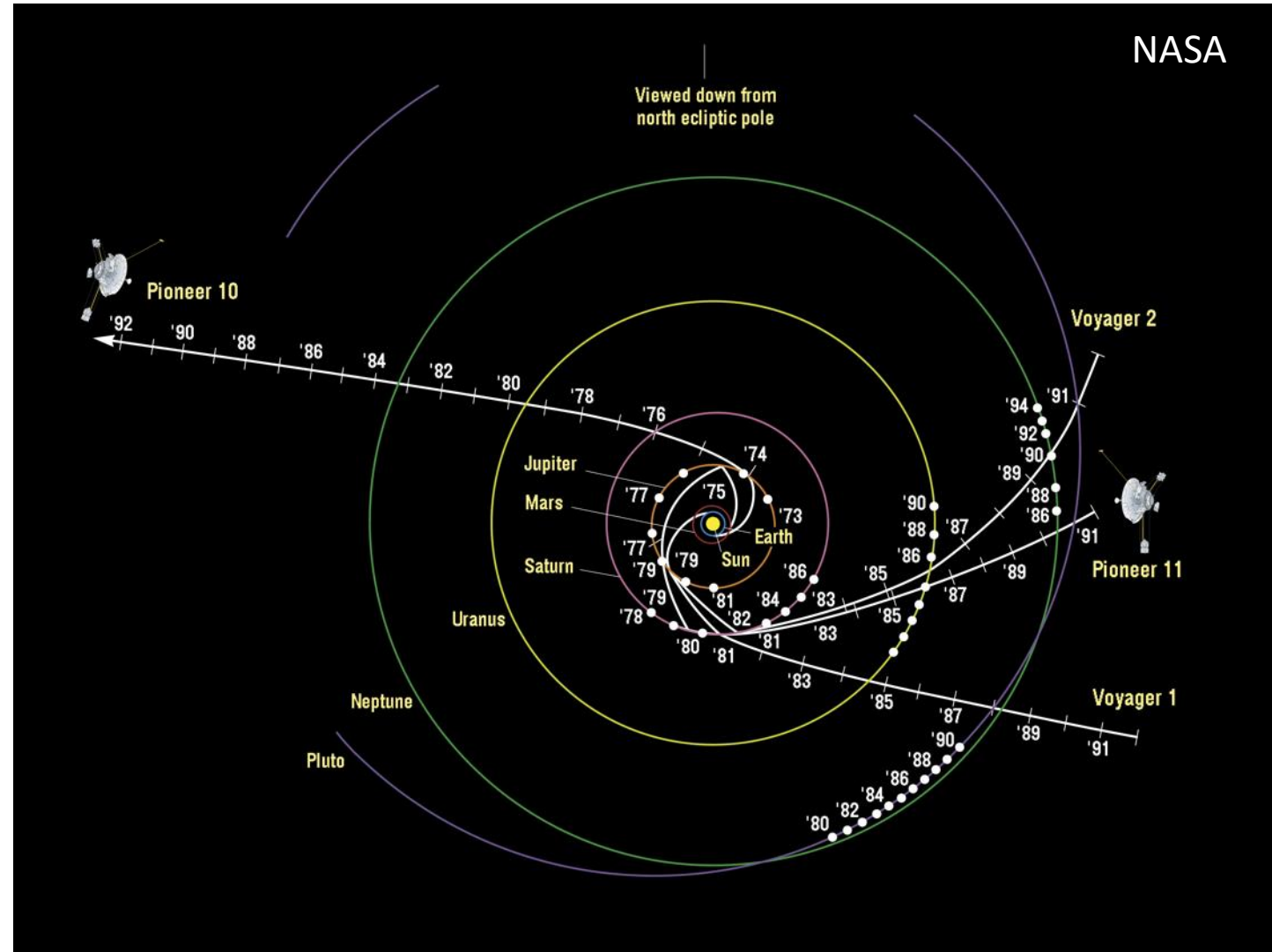
Central force from point mass → Keplerian motion

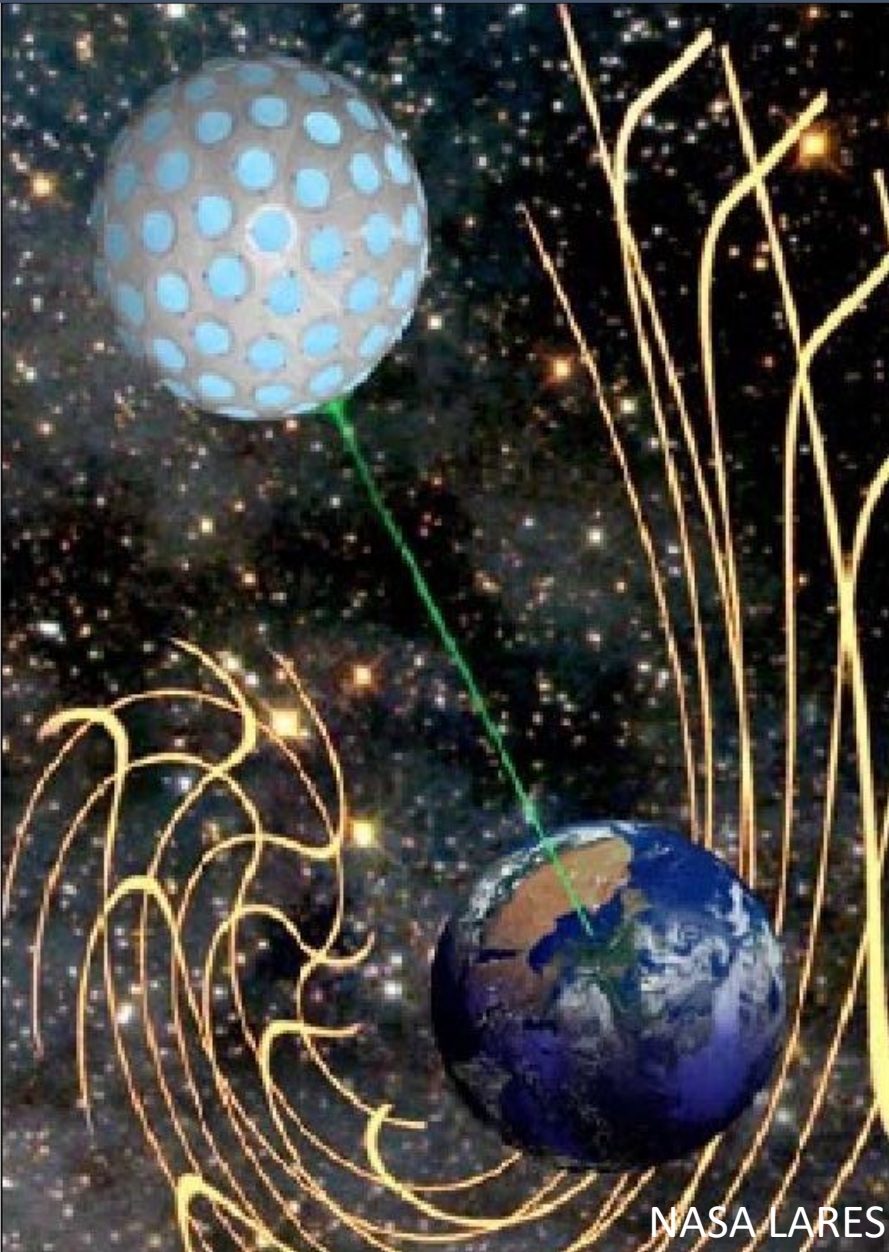
Approximation!

e.g. Pioneer anomaly

Observed in reality:
many other effects.

- Physical modelling of forces?
- How to solve EoM?
- Qualitative effects?
- Use in Mission planning?





Effects beyond pure central force:

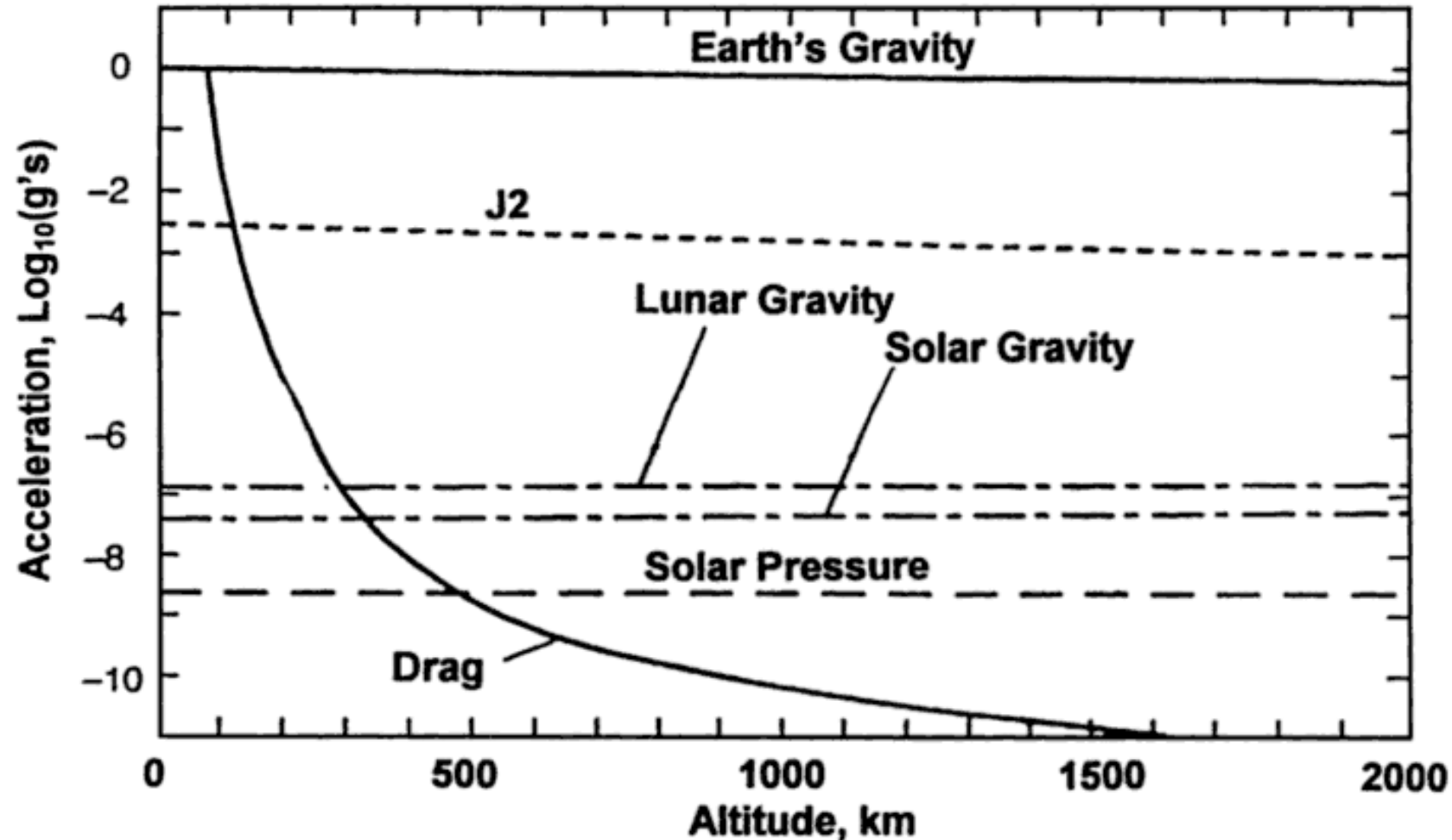
- Aerodynamics drag
- Solar radiation pressure
- Gravitational perturbation (non-spherical central body)
- Third body perturbations (Moon, Sun, ...)
- E&M drag
- Relativistic effects
- YORP effect
- Outgassing / leaking / low thrust
- ...

Relative strength of perturbations

Relative strength mostly depends on type of S/C and Altitude (Location)

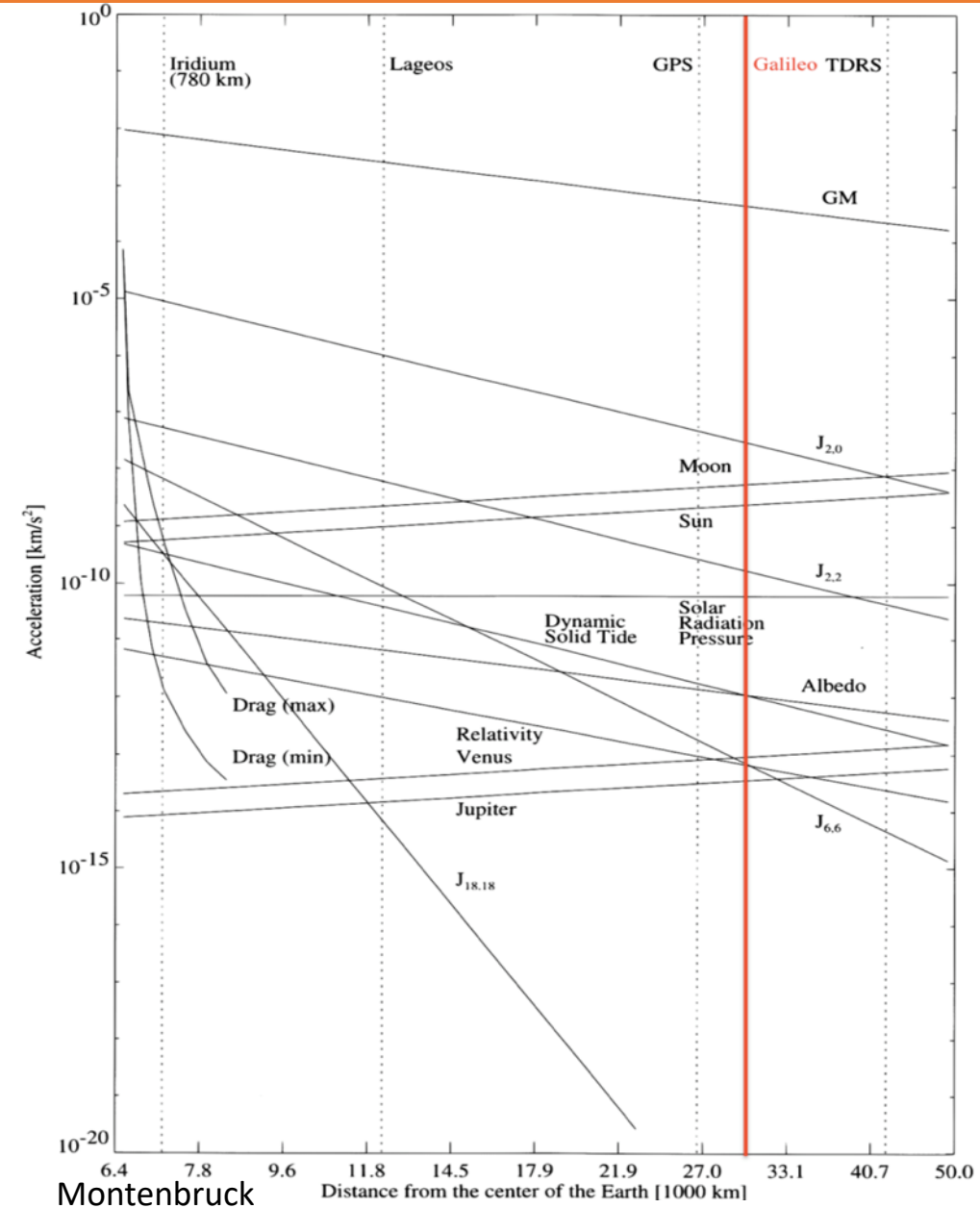
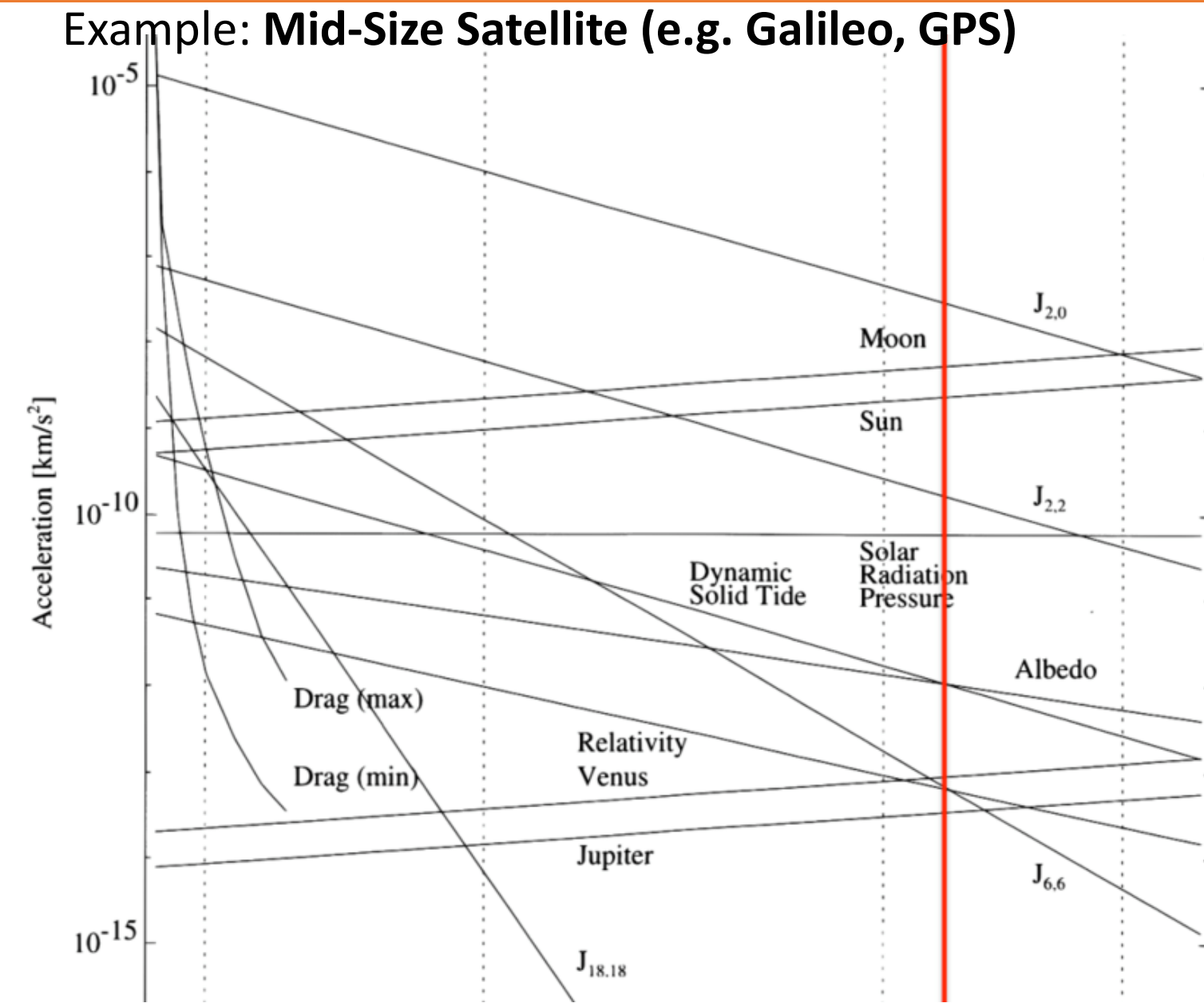
Example: **2U CubeSat (BeEagleSat)**

Cagri Kilic



Relative strength of perturbations

Example: Mid-Size Satellite (e.g. Galileo, GPS)



Main Orbit Perturbations

- Interaction between atmosphere and spacecraft
- Earth atmosphere:
 - extends up to about 500 km
 - most relevant below 150 km
- below: reentry dynamics / atmospheric flight (c.f. Adv. Astro, Flight Mech.)
- S/C design parameter: area to mass ratio A/m
- Lowers apogee of elliptic orbits → circularization

Requires model of interaction (physics) + model of atmosphere (density)

- Interaction between atmosphere and spacecraft
- Earth atmosphere:
 - extends up to about 500 km
 - most relevant below 150 km
- below: reentry dynamics / atmospheric heating (see v. Astro, Flight Mech.)
- S/C design parameter: area to mass ratio
- Lowers apogee of elliptical orbits → circularization

Requires model of drag force (physics) + model of atmosphere (density)

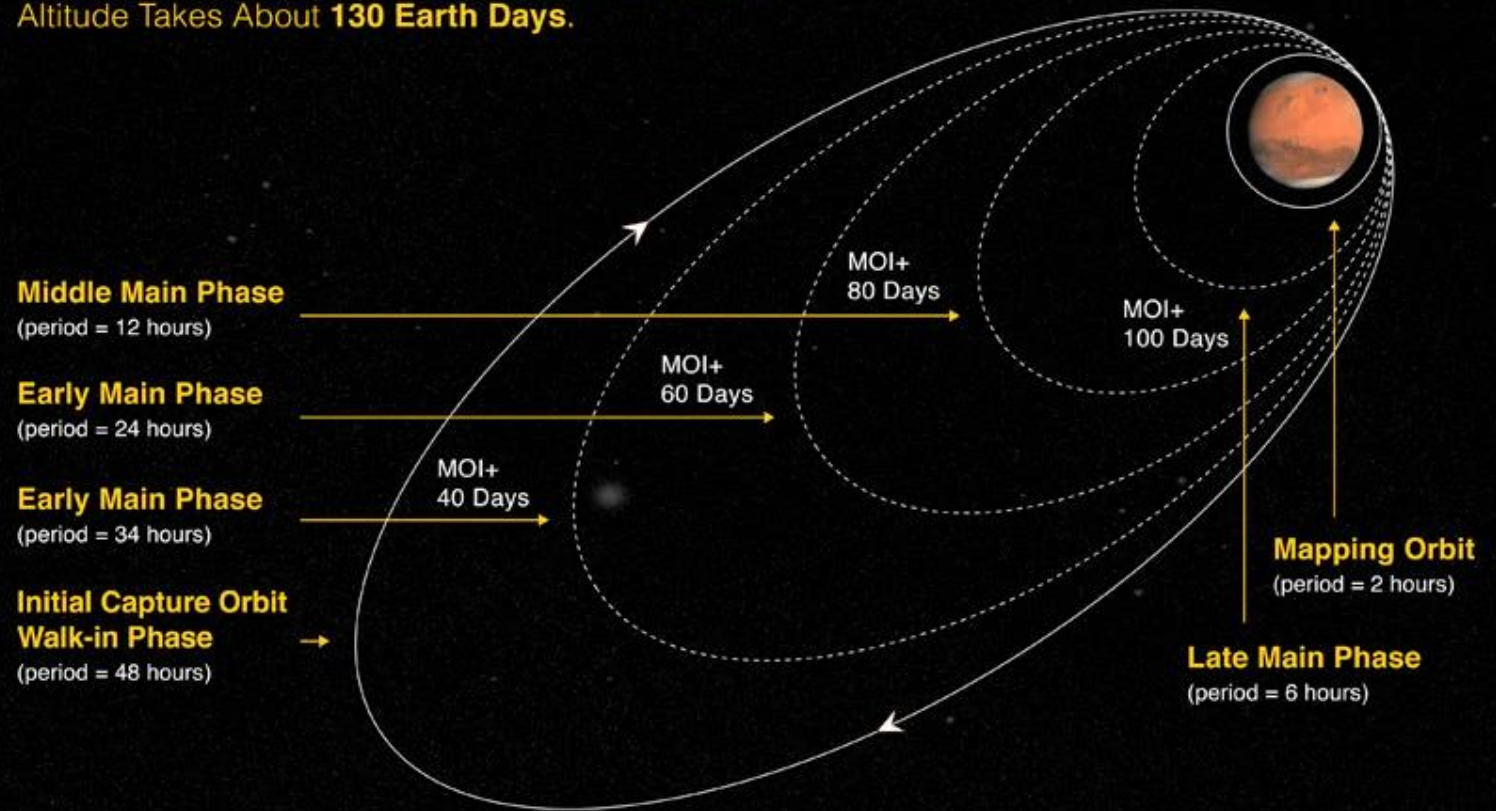
see Chapter 3 Video Lecture 7

- Automatic deorbiting
Life-cycle management
- Active debris removal
- Aero-breaking for
orbit insertion / capture
(interplanetary
trajectories)

Mars Global Surveyor Project

MGS AEROBRAKE BASELINE PROFILE

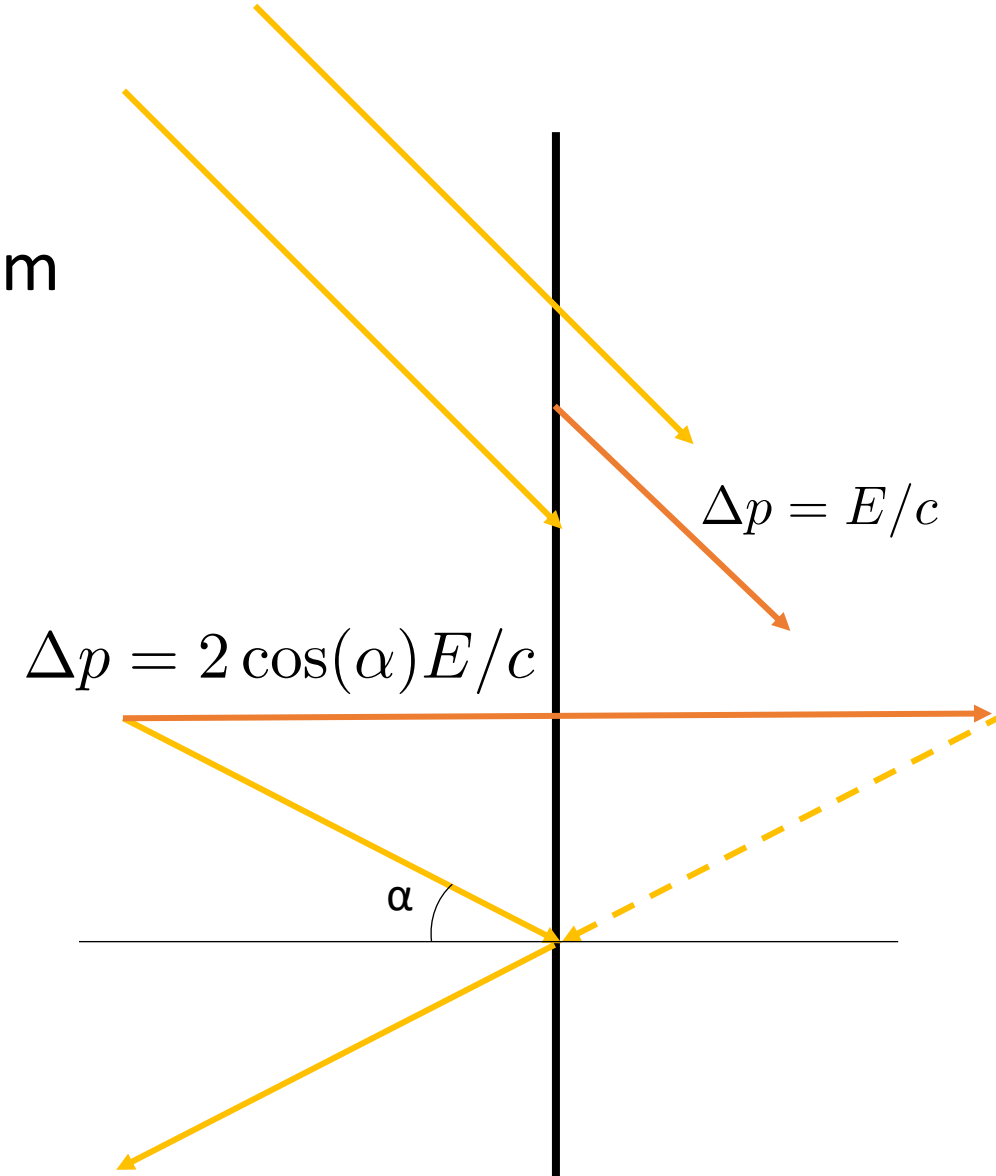
Aerobraking From Capture Orbit to Mapping Orbit
Altitude Takes About **130 Earth Days**.



- Pressure from photons
- Low thrust (@ Earth orbit)
 - $4.54 \mu\text{N}/\text{m}^2$ perfect absorber
 - $9.08 \mu\text{N}/\text{m}^2$ perfect reflector
- Can be controlled → steering, solar sailing
 - Orientation of sail
 - Reflectivity (new electrochromic materials)
- Still technology demonstrators
 - Structurally extremely challenging
 - Huge size yet light weight required

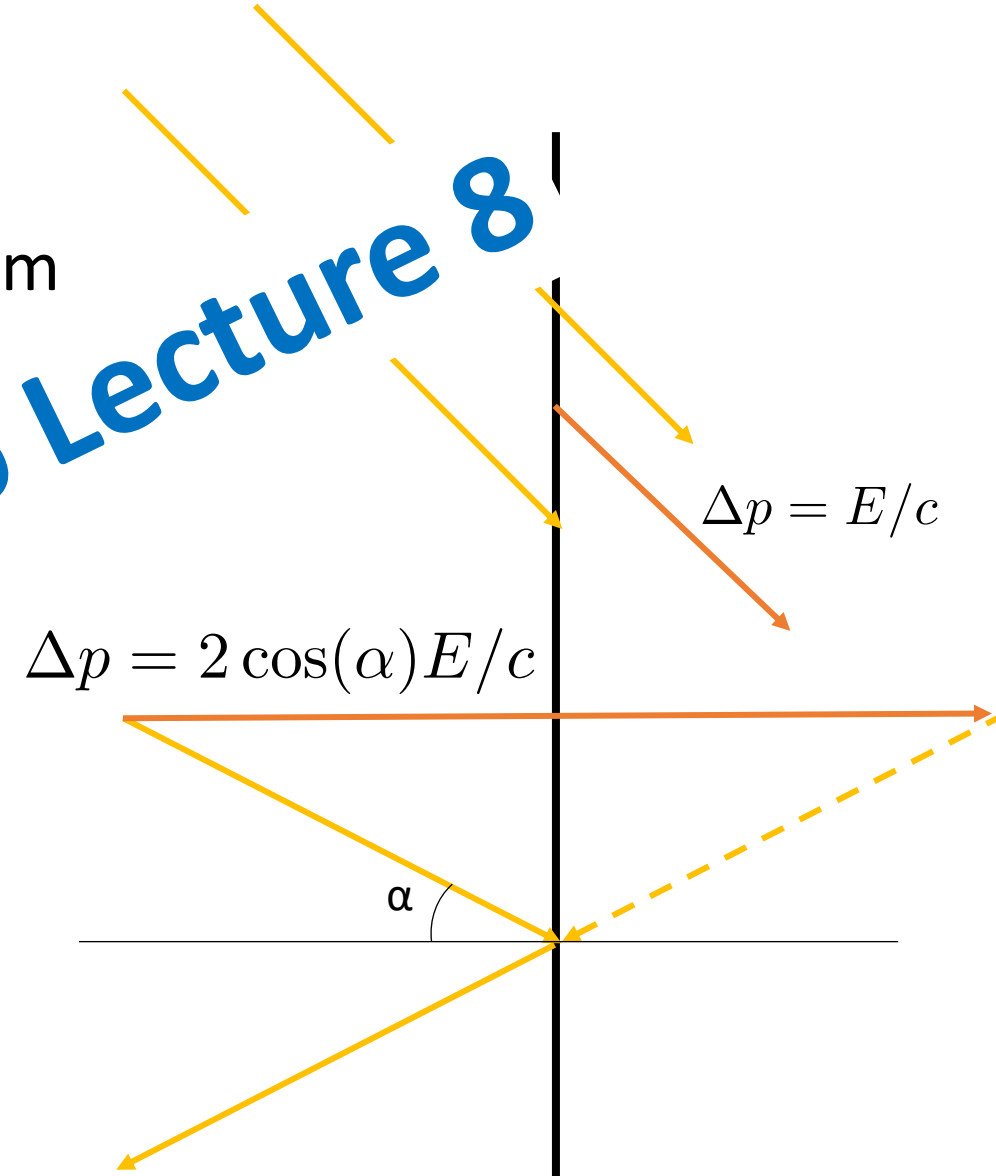


- Photons carry momentum $p = E/c$
- Absorbing / Reflecting light transfers momentum
- Direction:
 - absorbing: along light direction
 - reflecting: normal
 - reality: mix
- Depends on:
 - satellite attitude & geometry (area to mass A/m)
 - material transparency/reflectivity
 - incidence angle α

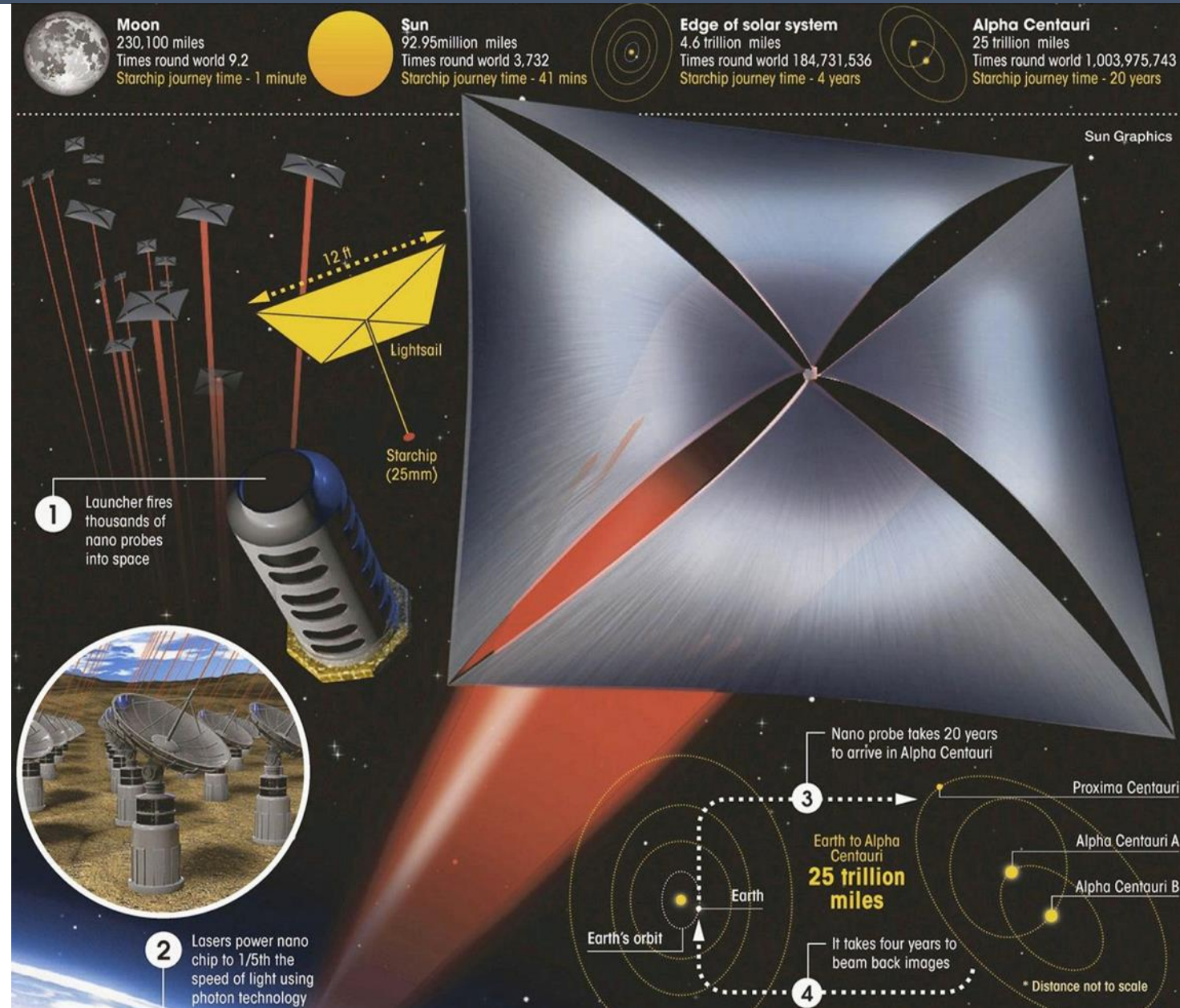


- Photons carry momentum $p = E/c$
- Absorbing / Reflecting light transfers momentum
- Direction:
 - absorbing: along light direction
 - reflecting: normal
 - reality: mix
- Depends on:
 - satellite attitude, (area to mass A/m)
 - material, reflectivity
 - incidence angle α

see Chapter 3 Video Lecture 8

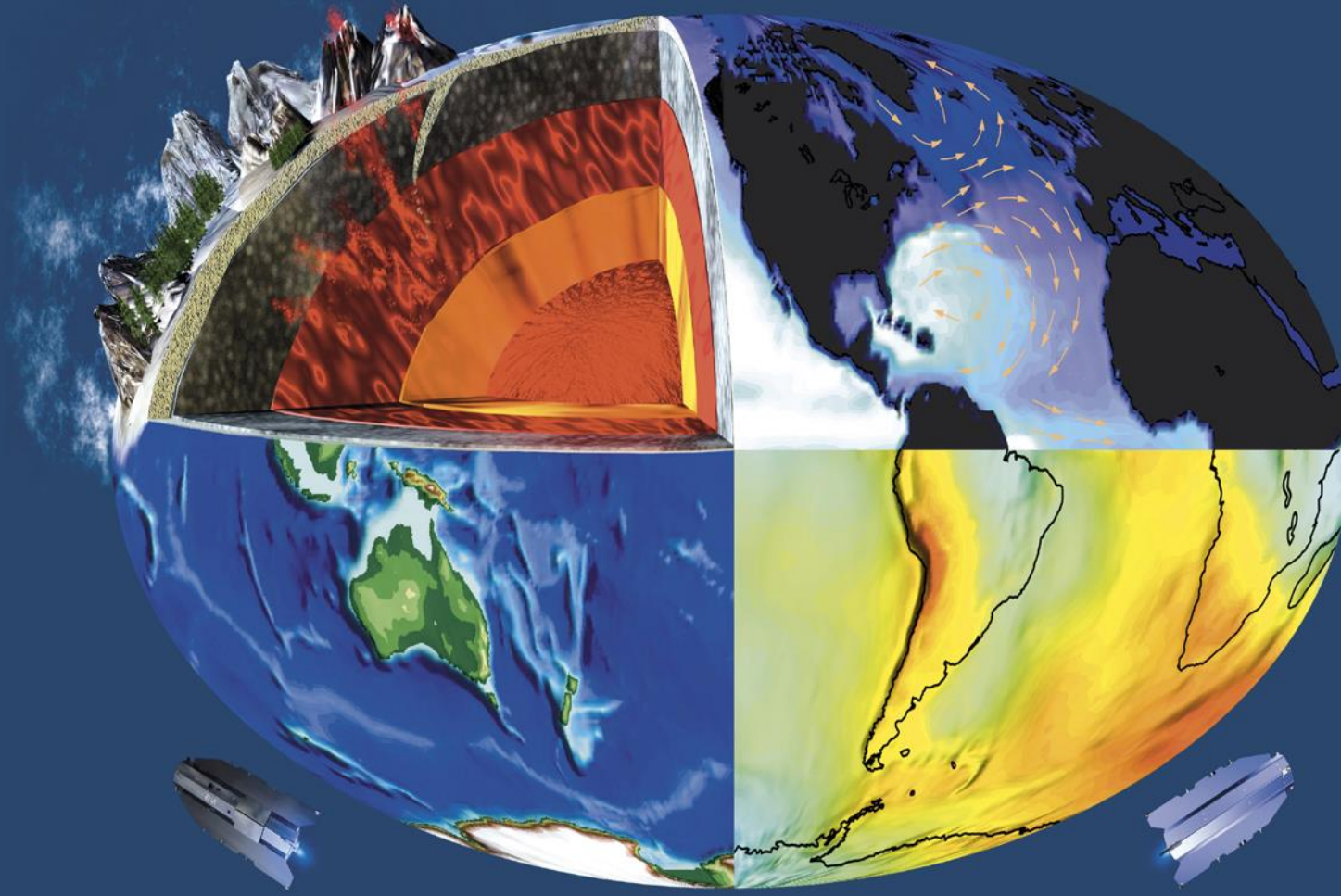


- Pole sitter orbits
- Active / passive debris removal and deorbiting
- Interstellar travel
([Breakthrough Starshot](#)):
chipsat + Earth based
high powered laser
to travel to Alpha Centauri



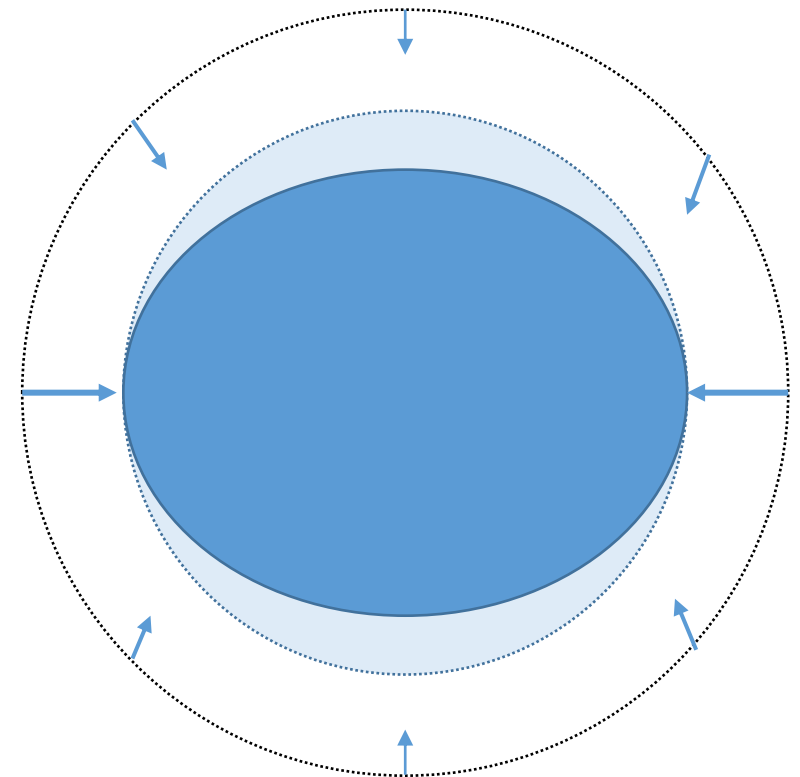
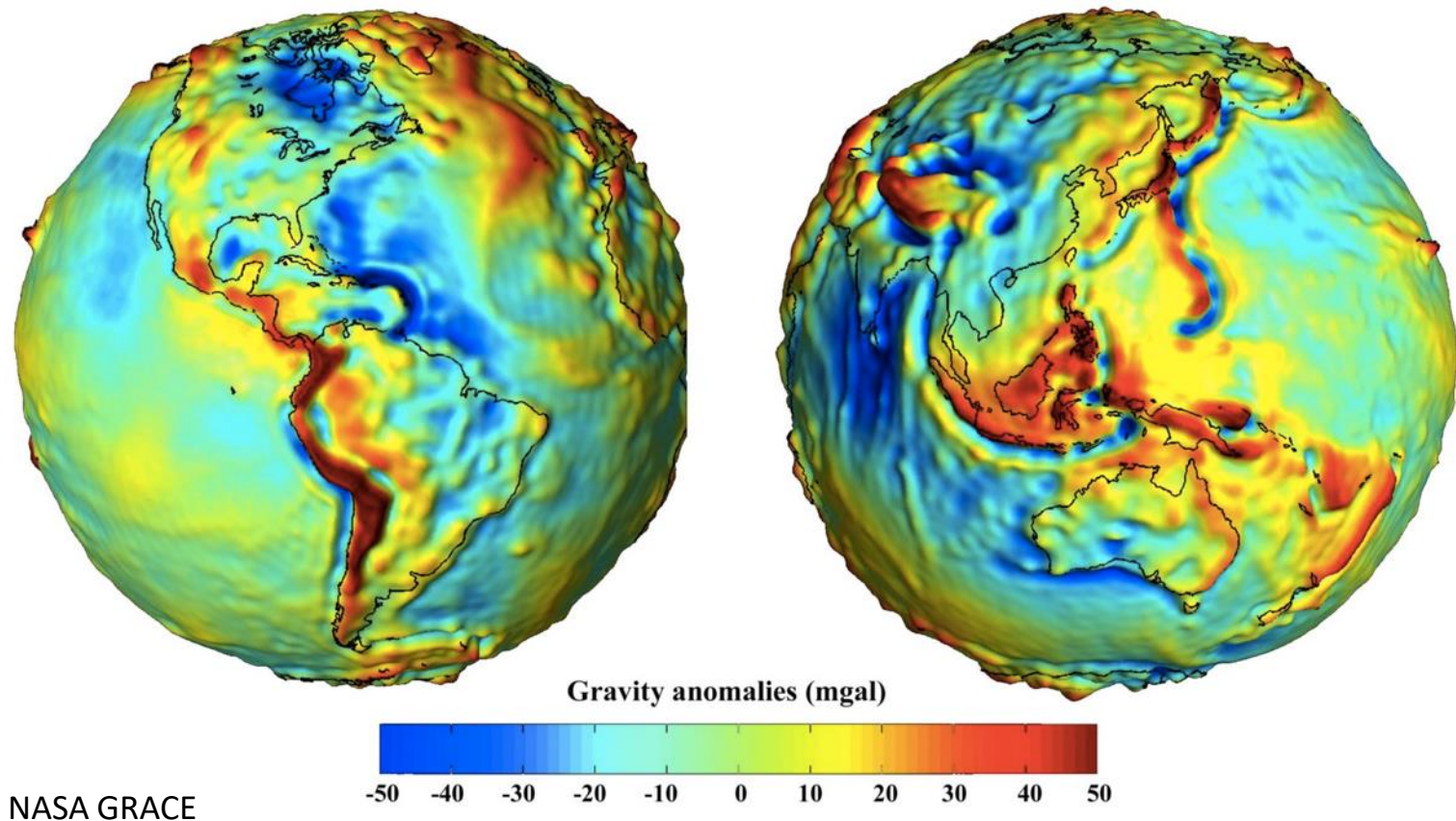
Earth gravitational field

ESA GOCE



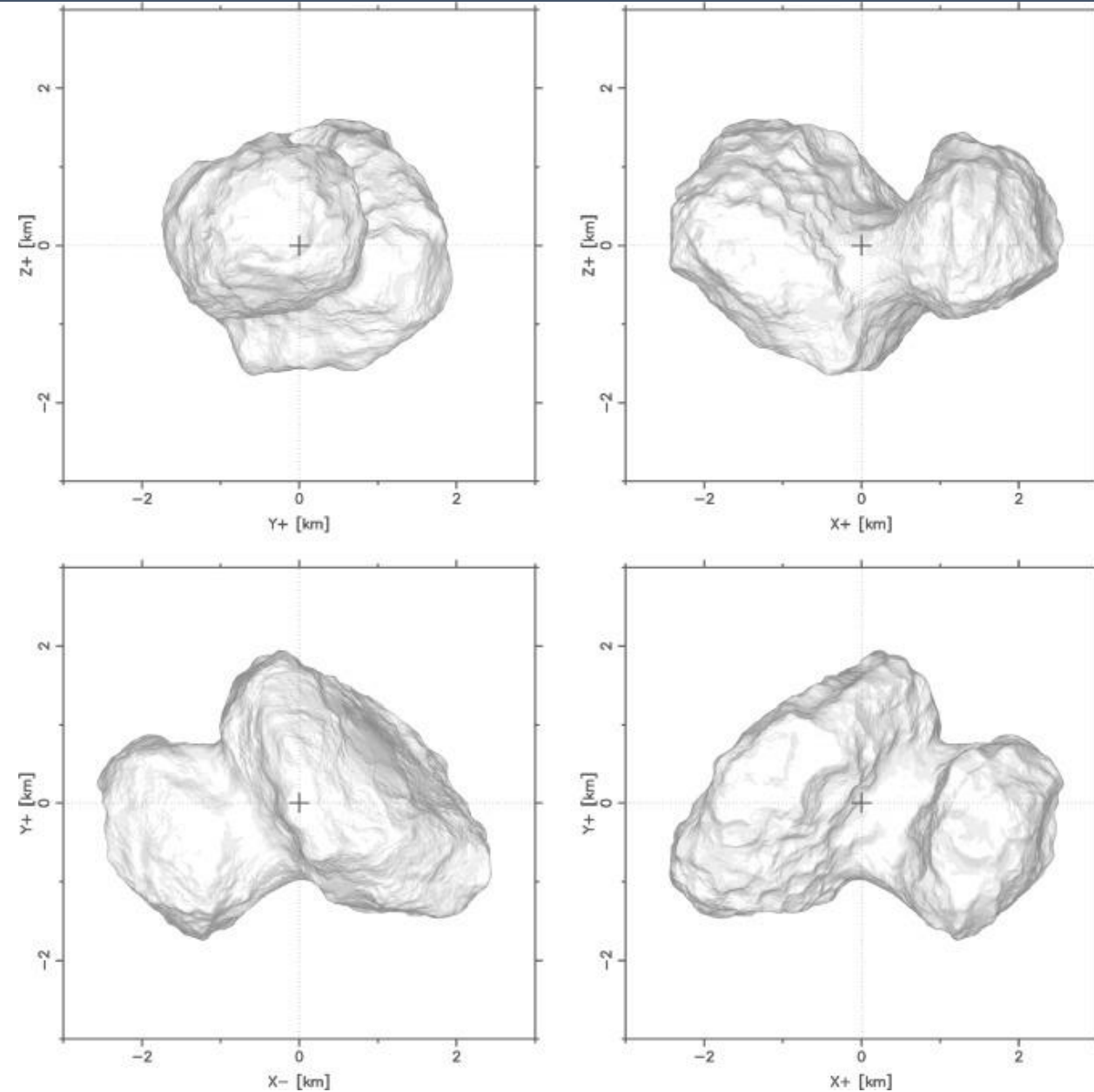
Shape of central body:

- Perfectly uniform sphere: can be treated as point mass
- Perfectly spherical shells of different density: still like point mass



Mesh grid model

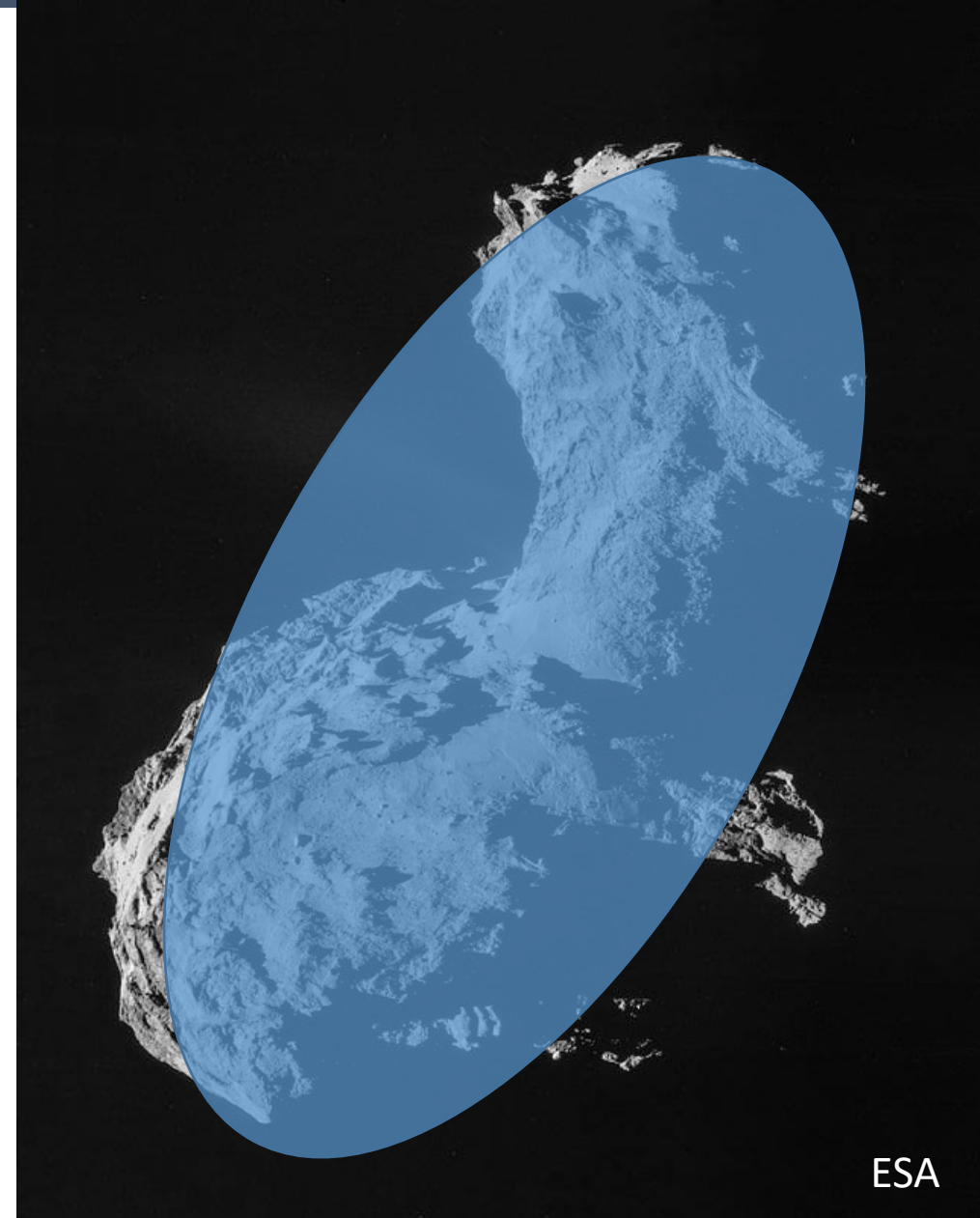
- From high-res imagery
- For very irregular bodies
 - High fidelity
 - Numerical
 - Computationally intensive



L. Jorda et al.: The global shape, density and rotation of Comet 67P/Churyumov-Gerasimenko from preperihelion Rosetta/OSIRIS observations
[Icarus 277](#), October 2016, Pages 257-278

Surrogate model

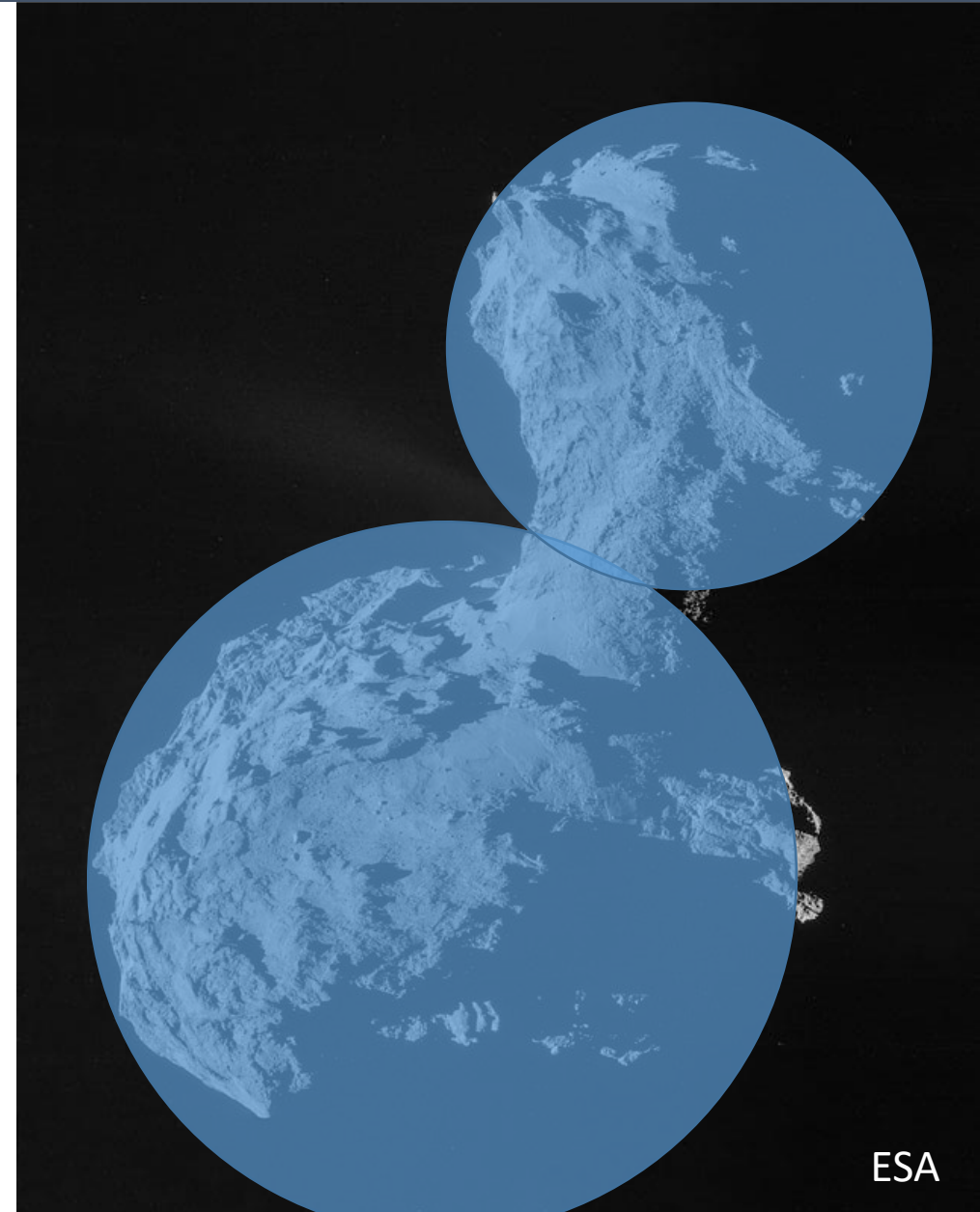
- From low-res observations
- When not much is known
 - Low fidelity
 - Analytical or Numerical
 - Matching of optical/radar observations



ESA

Surrogate model

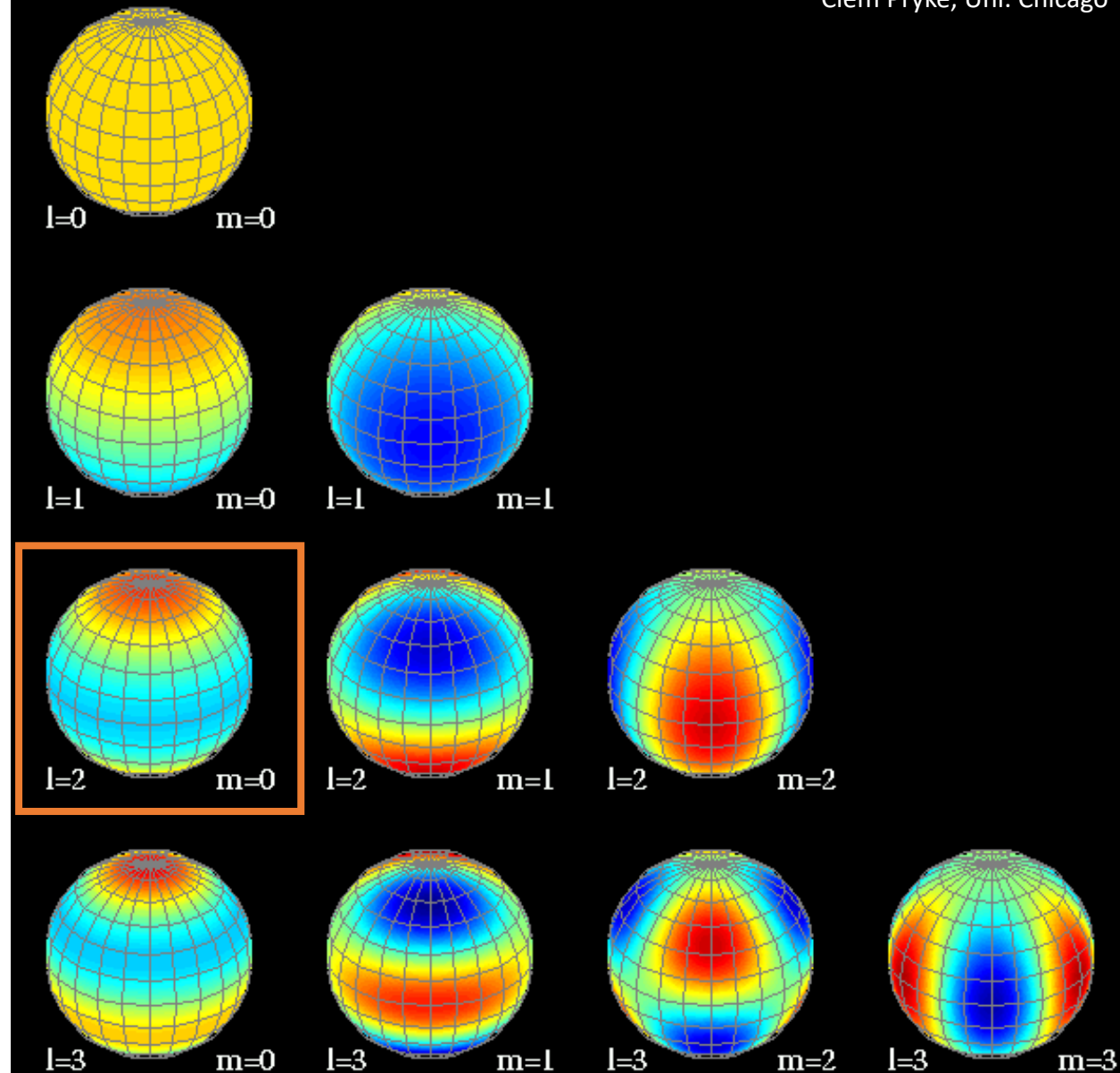
- From low-res observations
- When not much is known
 - Low fidelity
 - Analytical or Numerical
 - Matching of optical/radar observations



ESA

Spherical Harmonics

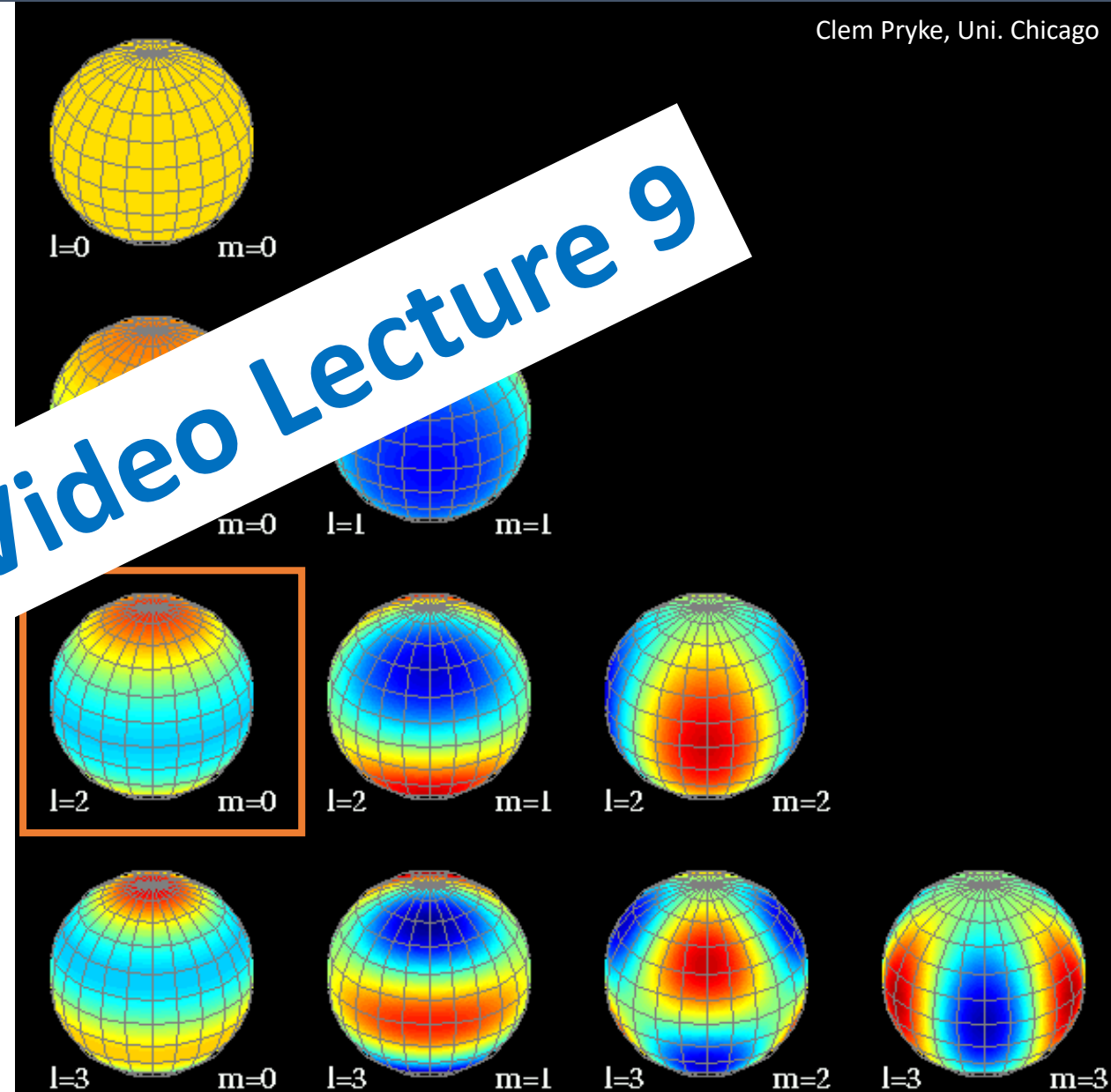
- From high-precision measurements
- Expansion of Potential
 - High fidelity for near-spherical bodies
 - Analytical
 - Perturbation from perfect sphere
- Most useful for planets
- Like JPEG compression algorithm



Spherical Harmonics

- From high-precision measurements
- Expansion of Potential
 - High fidelity for near-spherical bodies
 - Analytical
 - Perturbation from perfect
- Most useful
- Like JPEG compression algorithm

see Chapter 3 Video Lecture 9



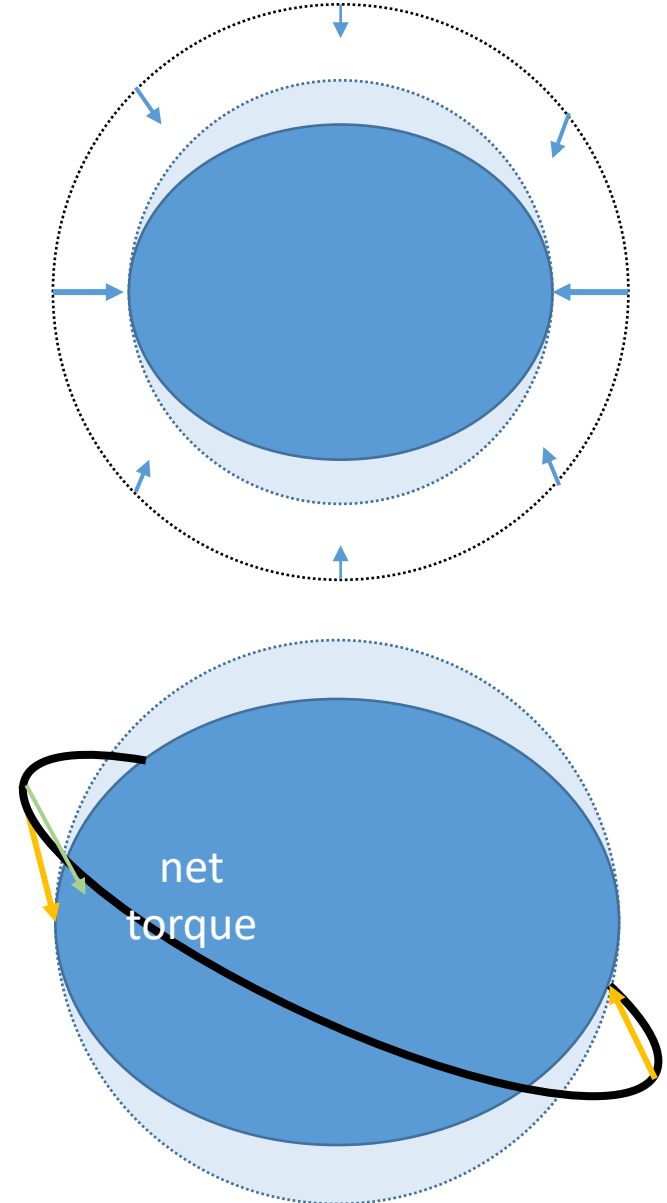
Coefficient sizes for Earth: $1 \cong 10^3 J_2 \cong 10^6 C_{l,m}; 10^6 S_{l,m}$

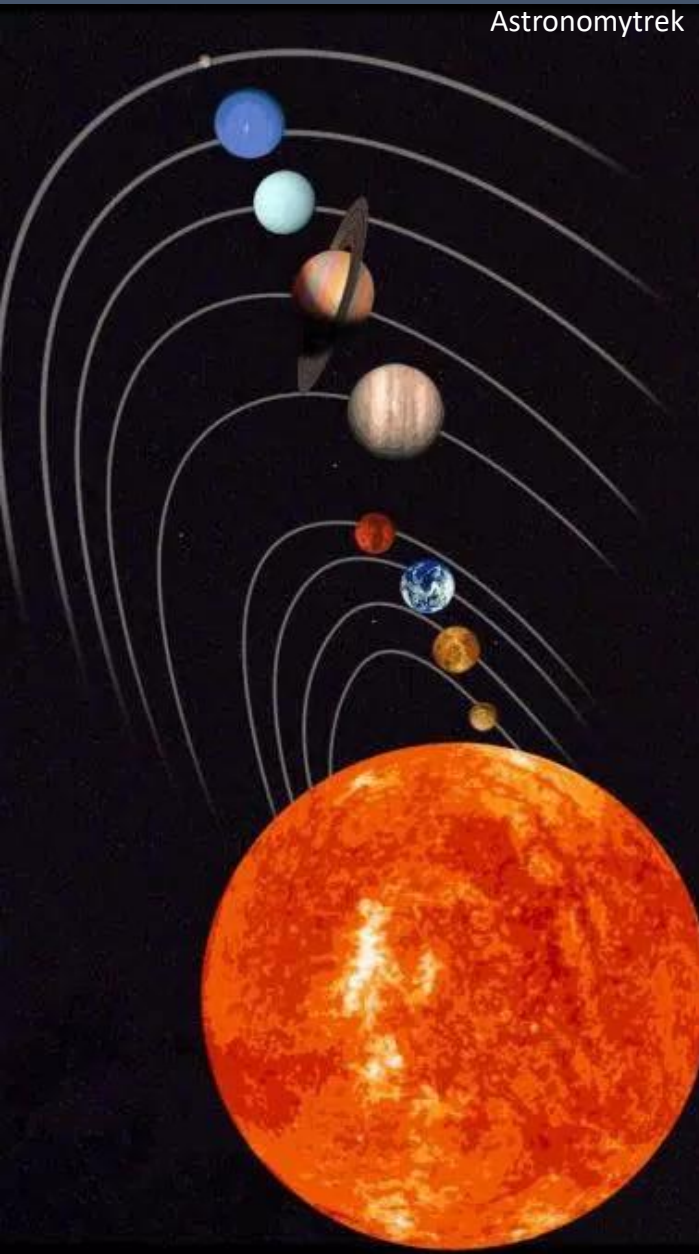
Effect:

- Describes equatorial bulge
- Produces net torque on orbits
- Rotation of orbital plane:
precession of line of nodes (RAAN)

Applications:

- Sun synchronous orbits (rotate with Earth)





- Other celestial bodies have influence as well
- For Earth orbiting satellites: Moon, Sun
Ephemerides:
 - Circular orbits
 - Keplerian orbits
 - Keplerian orbits with secular drift as polynomial
- Interplanetary: all planets
Ephemerides:
 - Typically from precomputed tables (e.g. NASA/JPL)

Mathematical formulation

$$\vec{a}_{\text{third body}} = \mu_3 \left(\overset{\text{force on S/C}}{\frac{\vec{r}_3 - \vec{r}}{|\vec{r}_3 - \vec{r}|^3}} - \overset{\text{force on primary}}{\frac{\vec{r}_3}{|\vec{r}_3|^3}} \right)$$

where:

- μ_3 – gravitational constant of third body ($= Gm_3$)
- r, r_3 – position of S/C & third body **relative to primary**

Pitfall:

Acceleration is not just due to force of third body on spacecraft!

Third body acts on both spacecraft and primary, hence two terms!

Mathematical formulation

$$\vec{a}_{\text{third body}} = \mu_3 \left(\frac{\vec{r}_3 - \vec{r}}{|\vec{r}_3 - \vec{r}|^3} + \frac{\vec{r}}{r^3} \right)$$

force on S/C force on primary

where:

- μ_3 – gravitational constant of third body (Gm_3)
- $\vec{r}, \vec{r}_3$ – position of S/C and third body relative to primary

Pitfall:

Acceleration is not just due to force of third body on spacecraft!

Third body acts on both spacecraft and primary, hence two terms!

see Chapter 3 Video Lecture 10

Specialized Orbit Perturbations

- Conductor moving in magnetic field $\rightarrow$ eddy currents
- Energy is dissipated
- Satellite motion in Earth magnetic field
- Electro-magnetic tether proposed for deorbiting

KoonPhysics (youtube)



Newtonian laws of motion are only an approximation!

■ Special relativity

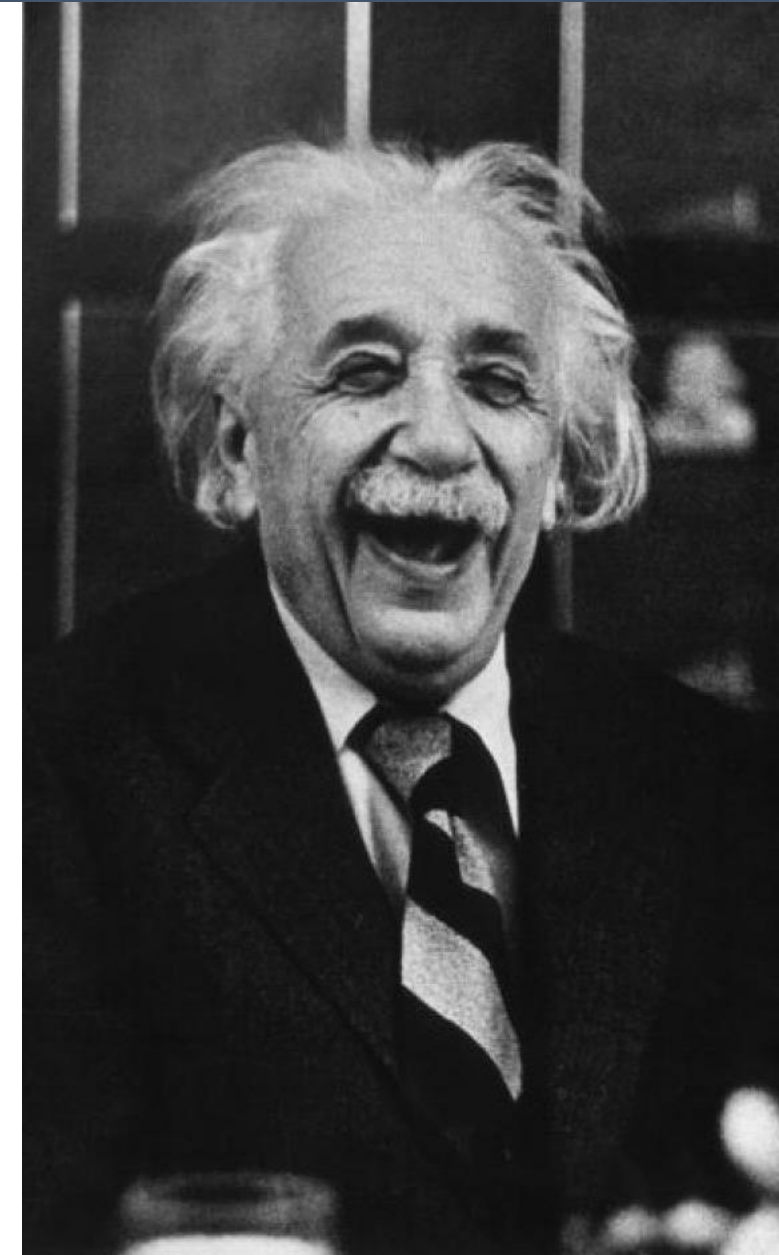
- Motion changes at high velocities (relative to speed of light)

■ General relativity

- All masses bend spacetime around them

$$\begin{aligned} \ddot{\mathbf{r}}_{i \text{ point mass}} = & \sum_{j \neq i} \frac{\mu_j (\mathbf{r}_j - \mathbf{r}_i)}{r_{ij}^3} \left\{ 1 - \frac{2(\beta + \gamma)}{c^2} \sum_{k \neq i} \frac{\mu_k}{r_{ik}} - \frac{2\beta - 1}{c^2} \sum_{k \neq j} \frac{\mu_k}{r_{jk}} \right. \\ & + \gamma \left(\frac{v_i}{c} \right)^2 + (1 + \gamma) \left(\frac{v_j}{c} \right)^2 - \frac{2(1 + \gamma)}{c^2} \dot{\mathbf{r}}_i \cdot \dot{\mathbf{r}}_j \\ & - \frac{3}{2c^2} \left[\frac{(\mathbf{r}_i - \mathbf{r}_j) \cdot \dot{\mathbf{r}}_j}{r_{ij}} \right]^2 + \frac{1}{2c^2} (\mathbf{r}_j - \mathbf{r}_i) \cdot \ddot{\mathbf{r}}_j \left. \vphantom{\sum_{j \neq i}} \right\} \\ & + \frac{1}{c^2} \sum_{j \neq i} \frac{\mu_j}{r_{ij}^3} \{ [\mathbf{r}_i - \mathbf{r}_j] \cdot [(2 + 2\gamma)\dot{\mathbf{r}}_i - (1 + 2\gamma)\dot{\mathbf{r}}_j] \} (\dot{\mathbf{r}}_i - \dot{\mathbf{r}}_j) \\ & + \frac{(3 + 4\gamma)}{2c^2} \sum_{j \neq i} \frac{\mu_j \ddot{\mathbf{r}}_j}{r_{ij}} + \sum_{m=1}^3 \frac{\mu_m (\mathbf{r}_m - \mathbf{r}_i)}{r_{im}^3} + \sum_{c,s,m} \mathbf{F} \end{aligned}$$

Estabrook, F.B.: "Derivation of Relativistic Lagrangian for n-Body Equations containing Relativity Parameters β and γ "
JPL IOM, Section 328, 14 June 1971.

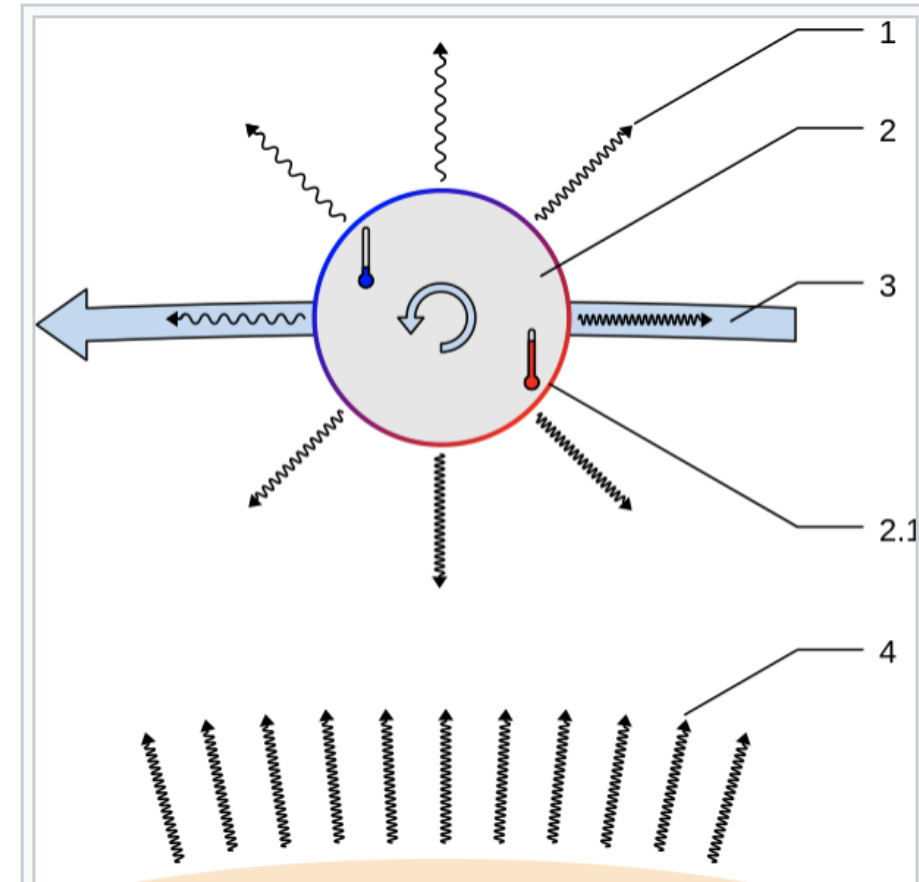


Yarkovsky effect

- Anisotropic heat distribution of rotating body leads to effective radiation pressure
- Causes perturbations on asteroids
- Hard to quantify exactly a priori, but measurable afterwards

Yarkovsky–O'Keefe–Radzievskii–Paddack (YORP)

- Different absorption of radiation results in net torque
- Increases rotation of asteroids
- Responsible for spin-up break-ups



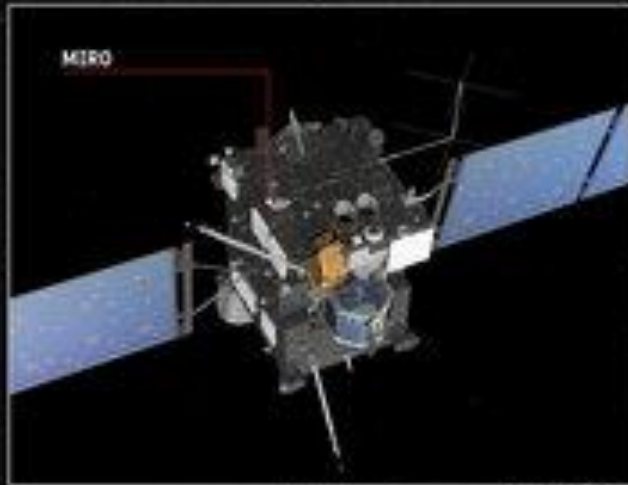
Yarkovsky effect:

1. Radiation from asteroid's surface
2. Prograde rotating asteroid
- 2.1 Location with "Afternoon"
3. Asteroid's orbit
4. Radiation from Sun

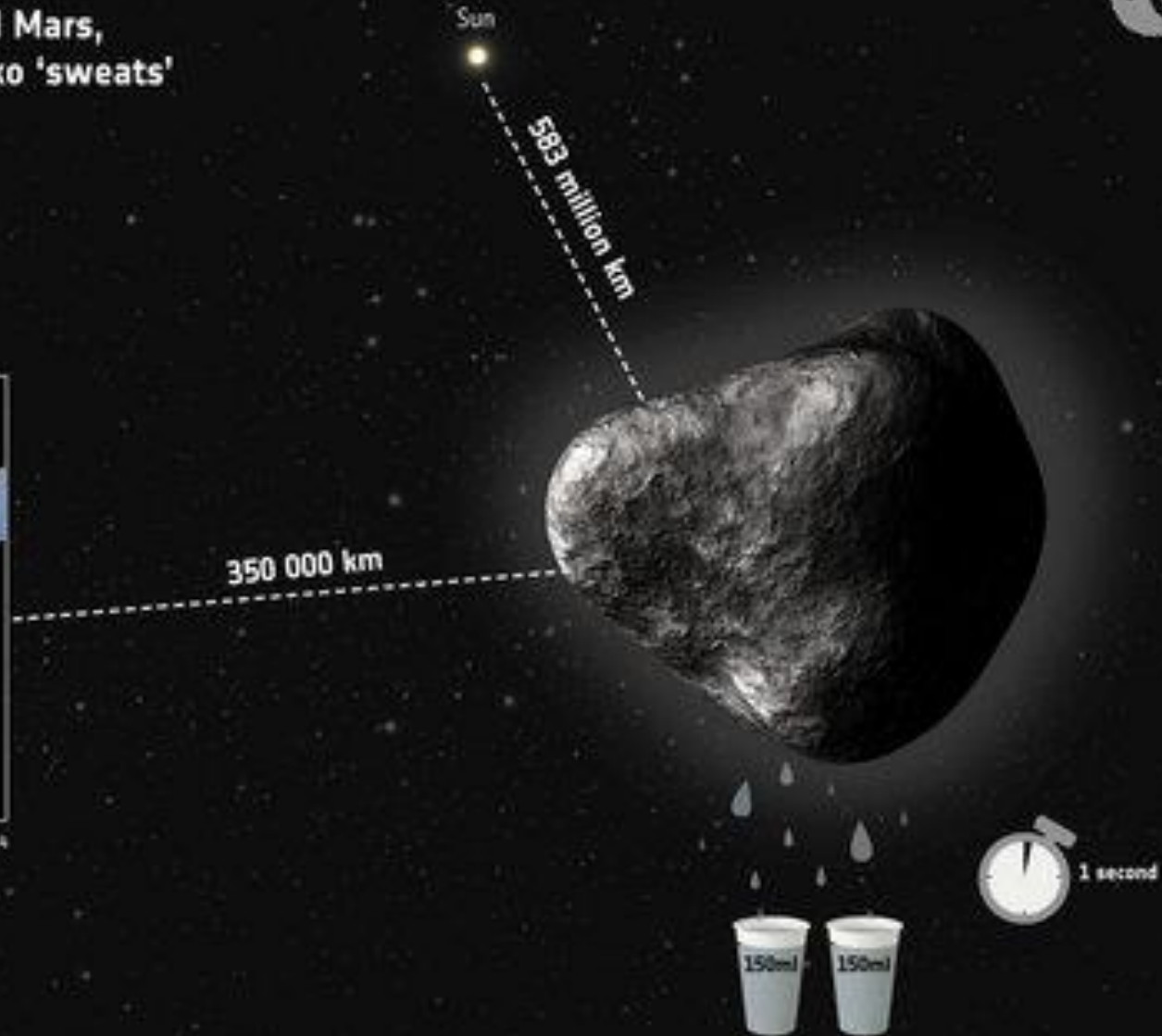
Wikipedia

Outgassing, Leaking, Low thrust

Even half way between Jupiter and Mars,
Comet 67P/Churyumov-Gerasimenko 'sweats'
up to 300ml of water a second



The observations were made on 6 June 2014





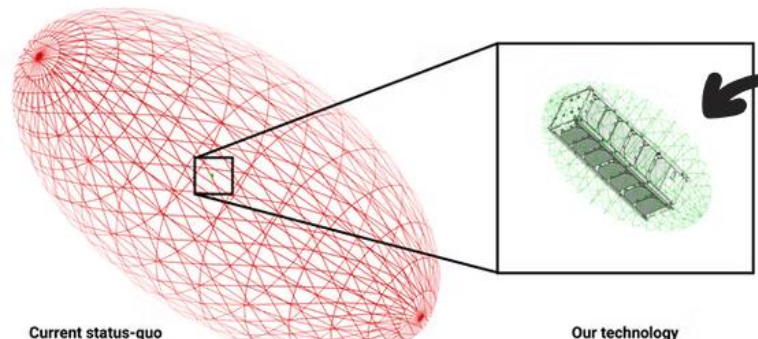
Data-driven Density Modelling (D3M)

- Real laser ranging satellite tracking data obtained from Lumi Space to increase the precision of density models for the upper atmosphere
- Big data mining techniques to continuously assimilate data into orbit prediction model
- Turn density model from external input into variable estimated based on real-time measurements through model-predictive control & machine learning
- Two main goals: identification and training of a mathematical representation of the density, and continuous update using real-time observations

➔ **10x more accurate orbit predictions**

Requirement: **First class Master's degree**

Topics: **Astroynamics, C++, ML, filtering & control**



Orbit Propagation Methods



Methods of orbit propagation:

- Analytical solutions
- Numerical methods
- Averaged (semi-analytical) methods

Special Perturbation theory:

- Perturbation written as forces
- Direct numerical propagation
- Good for visualization, high-accuracy numerical computation

General Perturbation theory:

- Perturbations written as forces or potentials
- Expanded in series
- Solved analytically or semi-analytically or numerically
- Good for analysis, understanding of effects, preliminary mission design

National Women's History Museum



Career:

- MS in Mathematics (Howard University, 1960)
- Army Map Service, US Census Bureau
- Computer (NASA, 1959)

Achievements:

- Oversaw Echo satellite tracking program (Goddard General Orbit Determination Program)
- Contributed to APL programming language
- Pioneered automatic documentation of computer code

Orbit Propagation Methods

Special Perturbation

Direct numerical integration:

$$\ddot{\vec{r}} = -\frac{\mu}{r^3}\vec{r} + \vec{a}_p$$

2nd order ODE in 3D

$$\begin{cases} \dot{\vec{r}} &= \vec{v} \\ \dot{\vec{v}} &= -\frac{\mu}{r^3}\vec{r} + \vec{a}_p \end{cases}$$

1st order ODE in 6D

+ Easy to implement
- Slow and/or inaccurate

- Integrate directly in Cartesian coordinates
- Use integration schemes such as Runge-Kutta, Adams-Moulton, BDF (scipy.integrate.solve_ivp, MATLAB ode45, C++ Boost.Numeric.Odeint)

Direct integration makes no use of analytical knowledge!

+ Faster, more accurate
- More difficult to implement

Encke's method:

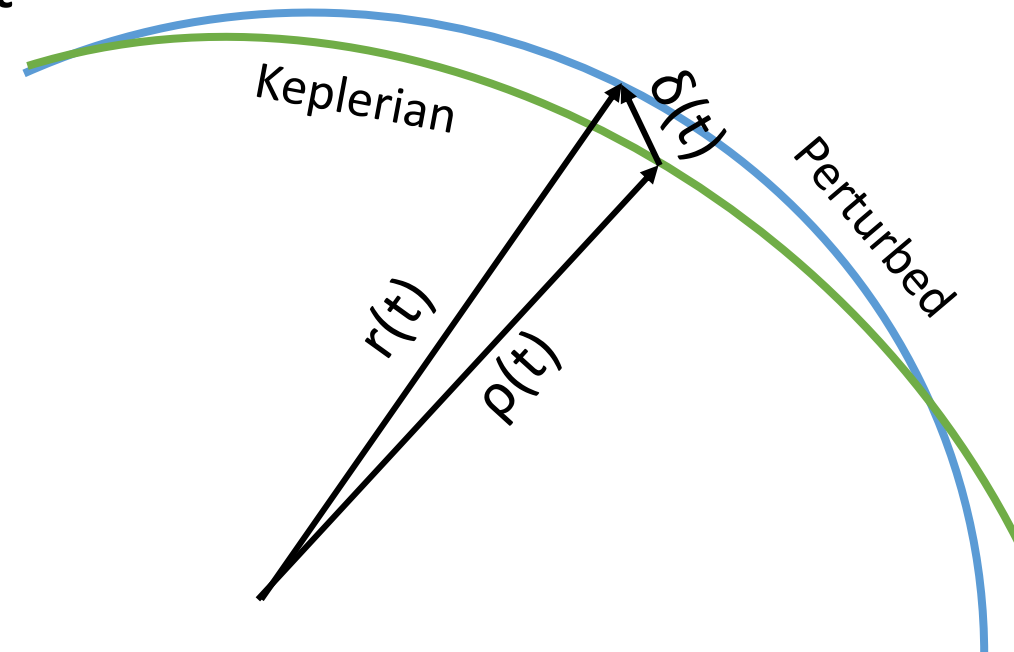
- Choose approximate Keplerian reference orbit
- Integrate only deviation numerically

$$\ddot{\delta} = \ddot{r} - \ddot{\rho}$$

$$= \mu \left(\frac{\vec{\rho}}{\rho^3} - \frac{\vec{r}}{r^3} \right) + \vec{a}_p$$

small

$$\ddot{\vec{r}} = \underbrace{-\frac{\mu}{r^3} \vec{r}}_{\substack{\text{large,} \\ \text{analytical}}} + \underbrace{\vec{a}_p}_{\substack{\text{small,} \\ \text{numerical}}}$$



Encke's method

Direct integration makes no use of analytical knowledge!

+ Faster, more accurate
- More difficult to implement

Encke's method:

- Choose approximate Keplerian reference
- Integrate only deviation number

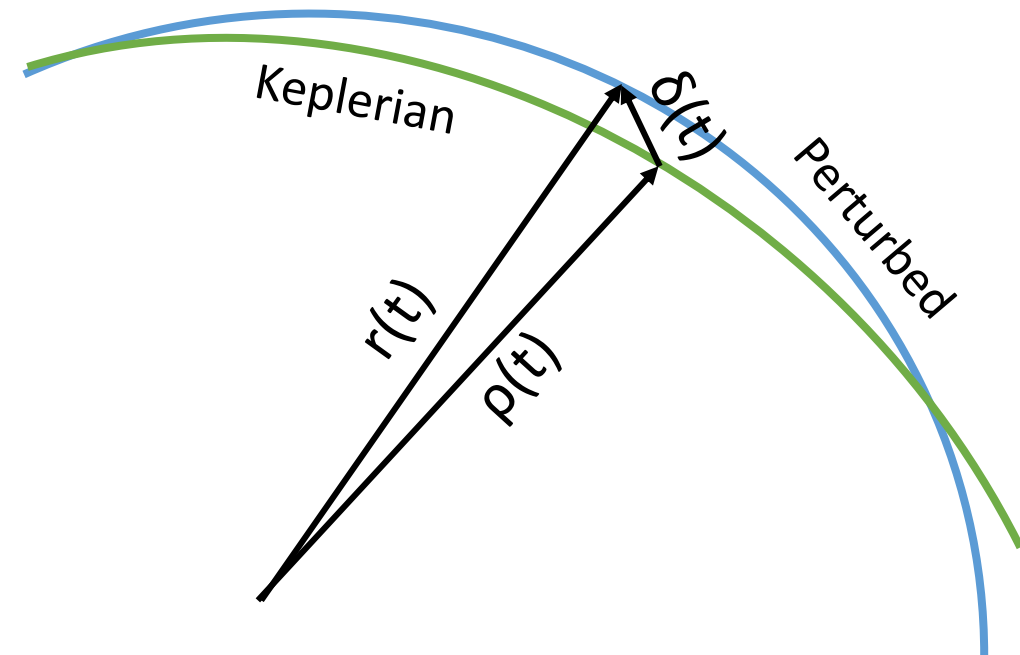
$$\ddot{\delta} = \ddot{\vec{r}} - \ddot{\vec{\rho}}$$

$$= \mu \left(\frac{\vec{r}}{r^3} - \frac{\vec{\rho}}{\rho^3} \right) + \vec{a}_p$$

small

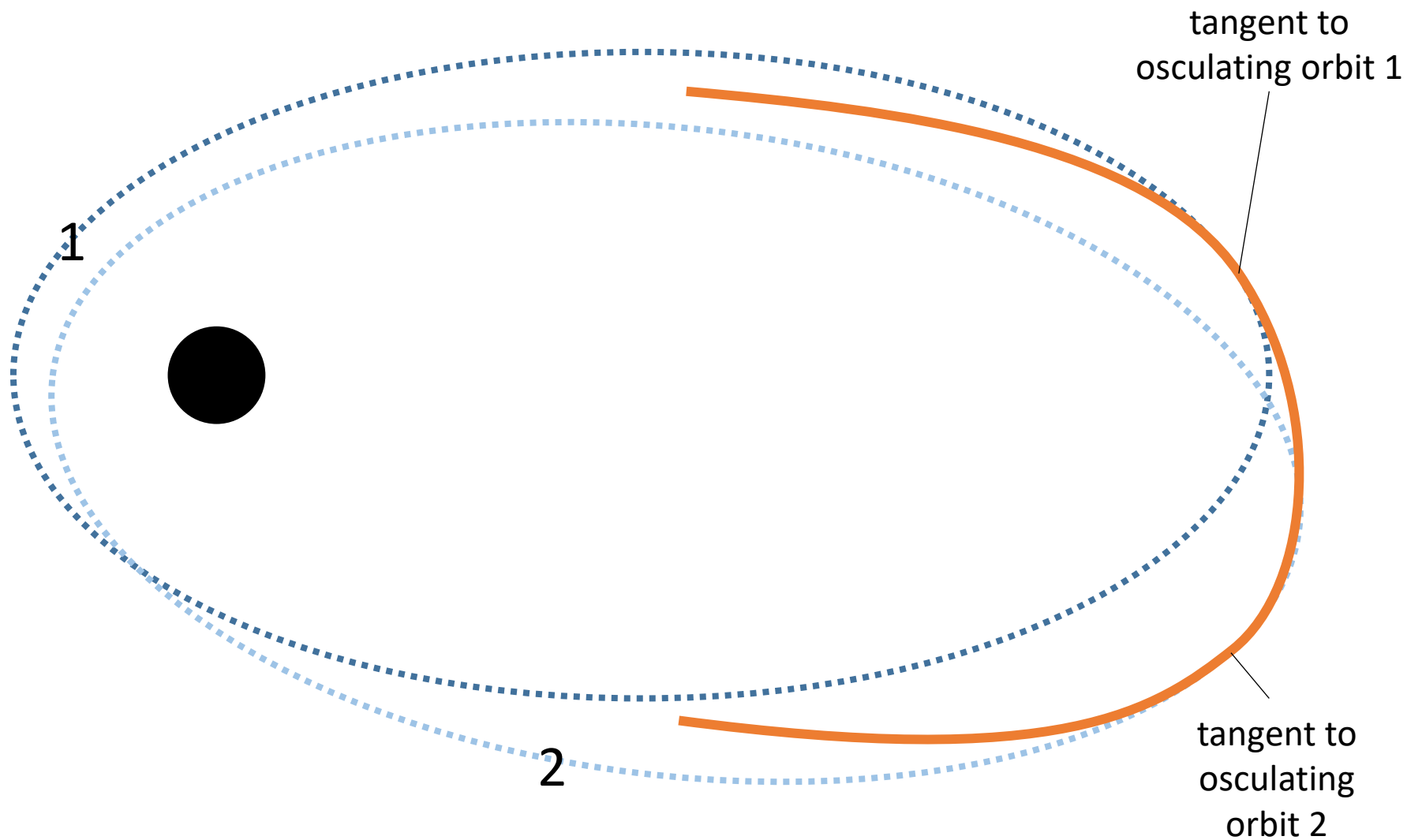
$$\ddot{\vec{r}} = \ddot{\vec{r}} + \vec{a}_p$$

large, analytical small, numerical



Orbit Propagation Methods

General Perturbation



Keplerian Motion:

$$\frac{da}{dt} = 0$$

$$\frac{de}{dt} = 0$$

$$\frac{di}{dt} = 0$$

$$\frac{d\Omega}{dt} = 0$$

$$\frac{d\omega}{dt} = 0$$

$$\frac{dM}{dt} = n$$

Equations of motion for osculating elements

$$\frac{da}{dt} = \frac{2e \sin \theta}{n\sqrt{1-e^2}} a_r + \frac{2a\sqrt{1-e^2}}{nr} a_s$$

$$\frac{de}{dt} = \frac{\sqrt{1-e^2} \sin \theta}{na} a_r + \frac{\sqrt{1-e^2}}{na^2 e} \left(\frac{a^2(1-e^2)}{r} - r \right) a_s$$

$$\frac{di}{dt} = \frac{r \cos(\theta + \omega)}{na^2 \sqrt{1-e^2}} a_w$$

$$\frac{d\Omega}{dt} = \frac{r \sin(\theta + \omega)}{na^2 \sqrt{1-e^2} \sin i} a_w$$

$$\frac{d\omega}{dt} = -\frac{\sqrt{1-e^2} \cos \theta}{nae} a_r + \frac{a(1-e^2)}{eh} \left[\sin \theta \left(1 + \frac{1}{1+e \cos \theta} \right) \right] a_s - \frac{r \cot i \sin(\theta + \omega)}{na^2 \sqrt{1-e^2}} a_w$$

$$\frac{dM}{dt} = n - \frac{1}{na} \left(\frac{2r}{a} - \frac{1-e^2}{e} \cos \theta \right) a_r - \frac{(1-e^2) \sin \theta}{nae} \left(1 + \frac{r}{a(1-e^2)} \right) a_s$$

- a_r, a_s, a_w – acceleration components in radial, horizontal and out of plane direction

Equations of motion for osculating elements

- a_r, a_s, a_w – acceleration components in radial, transverse and out of plane direction

$$\frac{da}{dt} = \frac{2e \sin \theta}{n\sqrt{1-e^2}} a_r + \frac{2a\sqrt{1-e^2}}{nr} a_s$$

$$\frac{de}{dt} = \frac{\sqrt{1-e^2} \sin \theta}{na} a_r + \frac{\sqrt{1-e^2}}{na^2 e} \left(\frac{a^2(1-e^2)}{r} - \dots \right)$$

$$\frac{di}{dt} = \frac{r \cos(\theta + \omega)}{na^2 \sqrt{1-e^2}} a_w$$

$$\frac{d\Omega}{dt} = \frac{r \sin(\theta + \omega)}{na^2 \sqrt{1-e^2}}$$

$$\frac{d\omega}{dt} = -\frac{\sqrt{1-e^2}}{na^2 e} + \frac{a(1-e^2)}{eh} \left[\sin \theta \left(1 + \frac{1}{1+e \cos \theta} \right) \right] a_s - \frac{r \cot i \sin(\theta + \omega)}{na^2 \sqrt{1-e^2}} a_w$$

$$\frac{dM}{dt} = n - \frac{1}{na} \left(\frac{2r}{a} - \frac{1-e^2}{e} \cos \theta \right) a_r - \frac{(1-e^2) \sin \theta}{nae} \left(1 + \frac{r}{a(1-e^2)} \right) a_s$$

see Chapter 3 Video Lecture 11

- Analytical

- Analytical solution to Gauss equations

$$\Omega(t) = \Omega(t_0) + \int_{t_0}^t \frac{d\Omega}{dt} dt \quad etc \dots$$

- Semi-Analytical

- Analytically average the effect of perturbations over one turn

$$\frac{d\bar{\Omega}}{dt} = \frac{1}{2\pi} \int_0^{2\pi} \frac{d\Omega}{dt} dM \quad etc \dots$$

- Numerical

- Integrate full Gauss equations directly

$$\dot{\Omega} = -\frac{3}{2} \left(\frac{J_2 \sqrt{\mu} R^2}{(1 - e^2)^2 a^{7/2}} \right) \cos i \quad [\text{rad/s}]$$

$$\dot{\omega} = -\frac{3}{2} \left(\frac{J_2 \sqrt{\mu} R^2}{(1 - e^2)^2 a^{7/2}} \right) \left(\frac{5}{2} \sin^2 i - 2 \right) \quad [\text{rad/s}]$$

where

- R – Earth radius (6378 km)
- J_2 – corresponding Earth J_2 (0.0010826)

Note:

Average rates of i , a , e due to J_2 are 0.

Application: Effect of Earth oblateness J_2

$$\dot{\Omega} = -\frac{3}{2} \left(\frac{J_2 \sqrt{\mu} R^2}{(1 - e^2)^2 a^{7/2}} \right) \cos i \quad [\text{rad/s}]$$

$$\dot{\omega} = -\frac{3}{2} \left(\frac{J_2 \sqrt{\mu} R^2}{(1 - e^2)^2 a^{7/2}} \right) \left(\frac{5}{2} \sin^2 i \right) \quad [\text{rad/s}]$$

where

- R – Earth radius (6378 km)
- J_2 – corresponding coefficient (0.0010826)

Note:

Average rates of i , a , e due to J_2 are 0.

see Chapter 3 Video Lecture 12

- Main Orbital Perturbations
 - Drag
 - SRP
 - Gravitational Field
 - Third body
- Additional Perturbations
- Relative Strengths / Relevance

- Osculating Elements
- Propagation Methods
 - Analytical
 - Semi-analytical
 - Numerical

Further reading

- Curtis, Chapter 10 (4th Edition)
- Vallado, Chapters 9 & 10

Orbital Mechanics (SESA6076)

Chapter 2: Orbit Perturbations & Propagation

Dr. Alexander Wittig

